Cost-Sensitive Customer Churn Prediction: A Business Utility-Driven Machine Learning Approach

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Abstract

This report presents a complete cost-sensitive churn prediction pipeline for the Telco Customer Churn benchmark dataset. We engineer 10 custom behavioral features, evaluate 9 classification models under SMOTE-balanced training with systematic 5-fold cross-validated hyperparameter optimization, and optimize decision thresholds against an asymmetric business utility function $U = 100 \cdot TP - 20 \cdot FP$, sweeping $\tau \in [0, 1]$ in steps of 0.01. AdaBoost achieves the highest net profit (20,040 monetary units at $\tau^* = 0.44$), while Logistic Regression ($\tau^* = 0.26$, 19,920) is selected as the final model for its superior interpretability. Post-hoc SHAP attribution identifies customer tenure, monthly charges, fiber optic service, month-to-month contracts, and electronic check payments as the top five churn drivers, directly validating patterns discovered during exploratory analysis. The final model is deployed to generate test predictions with 820 churn classifications at the optimized threshold.

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1 Introduction

Customer attrition (churn) directly threatens subscription-based business models. Telecommunications providers lose significant recurring revenue when subscribers terminate their contracts, yet retention campaigns are costly and must be targeted efficiently. This project addresses the challenge of identifying high-risk customers before they churn, enabling data-driven, cost-effective retention interventions.

The analysis follows a rigorous enterprise data science workflow: data ingestion and quality auditing, exploratory diagnostic analysis, domain-driven feature synthesis, class imbalance mitigation, multi-model classification with hyperparameter optimization, cost-sensitive threshold optimization, and post-hoc model interpretability. Crucially, this assignment moves beyond traditional symmetric statistical metrics to a business utility framework that balances the financial trade-offs of customer retention.

The Telco Customer Churn dataset comprises 7,043 subscriber records with 20 feature covariates organized into three functional dimensions:

- **Demographic Attributes:** gender, SeniorCitizen (binary 0/1), Partner, Dependents.
- **Service Architecture:** PhoneService, MultipleLines, InternetService (DSL, Fiber optic, No), OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies.
- **Account Financials:** tenure (months), Contract type (Month-to-month, One year, Two year), PaperlessBilling, PaymentMethod (Electronic check, Mailed check, Bank transfer, Credit card), MonthlyCharges, TotalCharges.

2 Exploratory Data Analysis

2.1 Structural Inspection and Data Cleaning

Programmatic inspection confirmed 7,043 rows and 21 columns. The `TotalCharges` column contained blank strings coerced into a generic object data type. After isolation and parsing via `pd.to_numeric(errors="coerce")`, 8 missing values were identified. Rather than applying a simple global median, we employed a **tenure-cohort-based conditional median imputation**: customers were grouped into six tenure bins ($[0, 12], (12, 24], (24, 36], (36, 48], (48, 60], (60, \infty)$ months) and missing values were replaced with the median of their respective cohort.

This strategy preserves the strong positive correlation between tenure and total charges ($\rho = 0.83$), introducing substantially less bias than a column-wide median imputation. A customer with 60+ months of tenure who happens to have a missing `TotalCharges` entry should not receive the median value of all customers – including those with just 1 month of tenure – as their estimated value.

2.2 Target Distribution and Class Imbalance

The target variable exhibits pronounced class asymmetry:

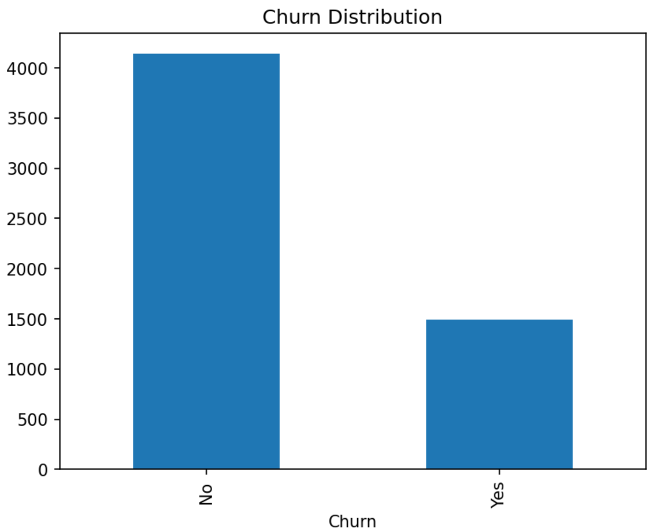

Figure 1: Target distribution: 73.5% non-churn, 26.5% churn (1,196 of 4,507 samples). This 3:1 ratio motivates synthetic oversampling to prevent majority-class bias.

The exact imbalance ratio is $\frac{1{,}196}{4{,}507} \approx 0.265$.

2.3 Univariate and Bivariate Analysis

Several key patterns emerged that directly inform feature engineering and modeling focus.

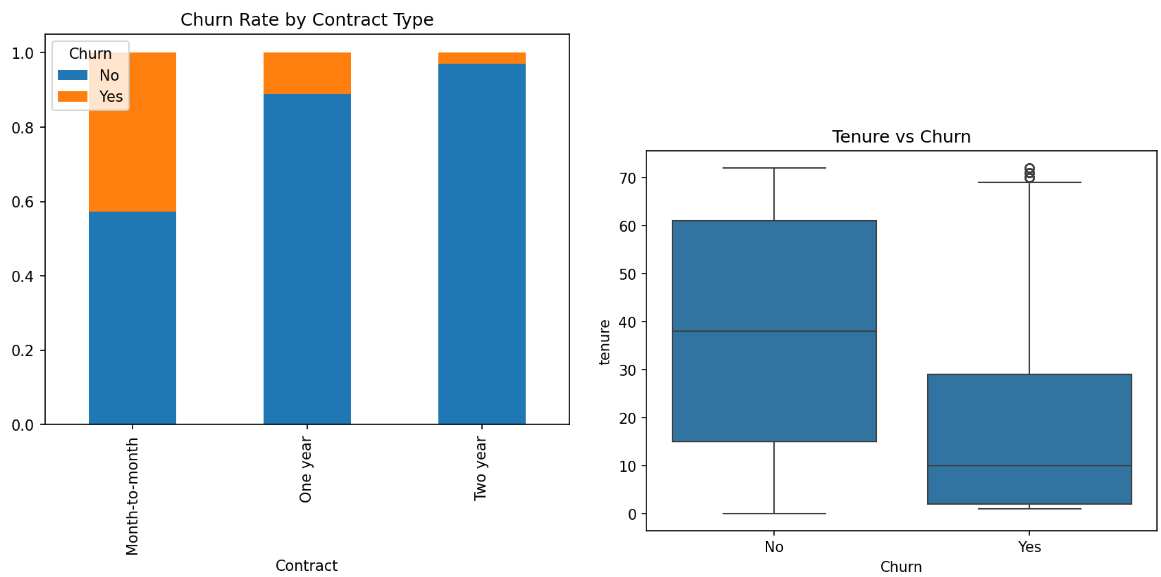

Figure 2: Left: Churn rate by contract type – month-to-month customers churn at 42.7%, versus 11.1% for one-year and 2.9% for two-year contracts. Right: Tenure distribution – churners have a median tenure of 10 months compared to 38 months for loyal customers. The first year of service is the highest-risk period.

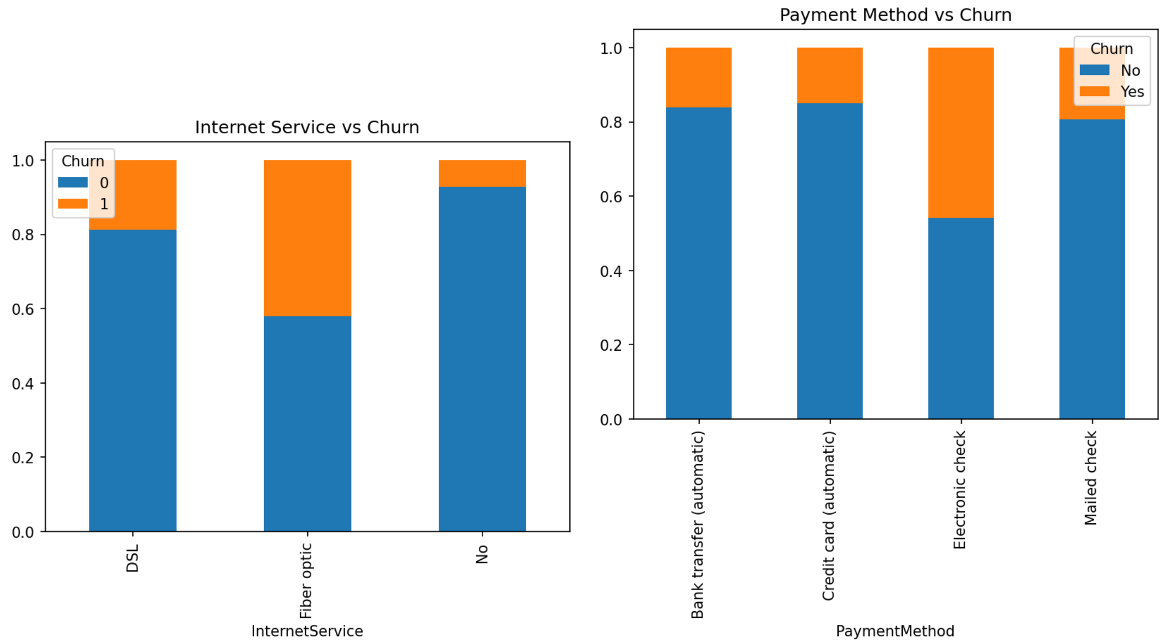

Figure 3: Left: Internet service vs churn – fiber optic users churn at 42.1%, more than double the DSL rate (18.7%). Right: Payment method vs churn – electronic check payers churn at 45.7%, far exceeding automatic payment methods (credit card: 15.2%, bank transfer: 15.0%, mailed check: 19.2%).

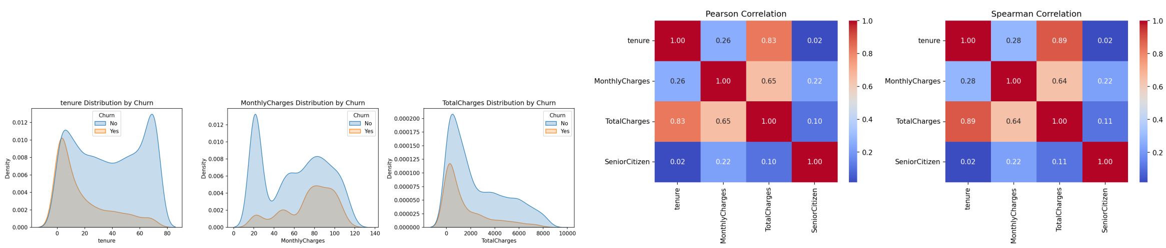

Figure 4: Left: KDE distributions of numeric features split by churn status. Tenure and MonthlyCharges show the largest separation. Right: Spearman correlation heatmap – the strongest correlation is between tenure and TotalCharges ($\rho = 0.83$).

2.4 Analytical Synthesis

The EDA identified the following high-risk customer profiles:

- **Early-tenure customers:** Median tenure of churners is 10 months versus 38 months for loyal customers.
- **Month-to-month contracts:** 42.7% churn – the single strongest categorical risk signal.
- **Fiber optic internet:** 42.1% churn despite being a premium product, suggesting pricing or reliability concerns.
- **Electronic check payment:** 45.7% churn, indicating payment friction as a measurable churn catalyst.

- **Higher monthly charges:** Churners average \$74.86/month versus \$61.34 for non-churners.

These findings directly guide the feature engineering strategy.

3 Feature Engineering

Ten custom features were synthesized, each grounded in a behavioral hypothesis explaining its expected predictive signal. All transformations are embedded within a `scikit-learn ColumnTransformer` inside an `ImbPipeline`, ensuring scaling and encoding parameters are learned exclusively from training folds to prevent data leakage.

Feature	Type	Behavioral Hypothesis
`TenureRange`	Ordinal	Binned tenure $([0, 12], (12, 24], \ldots, (60, \infty))$ captures non-linear churn risk.
`TotalServices`	Discrete	More services $\rightarrow$ higher switching costs $\rightarrow$ lower churn.
`Security`	Boolean	Joint OnlineSecurity + DeviceProtection signals security-conscious, engaged users.
`Entertainment`	Boolean	StreamingTV + StreamingMovies subscribers form entrenched habits.
`SeniorTechSupport`	Boolean	Seniors with TechSupport represent a high-dependency segment.
`BillingAndPayment`	Boolean	Paperless billing + electronic check flags operational friction.
`MonthlyChargesRange`	Ordinal	Fixed 20-unit bins $([0, 20], [20, 40], \ldots)$ capture tier-based pricing.
`AvgMonthlyPerService`	Continuous	Cost-per-service ratio; high values may indicate poor value perception.
`TenureContractRisk`	Boolean	Month-to-month + tenure ≤ 12 months: highest-risk segment.
`LoyaltyFactor`	Ordinal (1–5)	Composite score from contract type and tenure duration.

Table 1: Engineered features with types and behavioral hypotheses. All transformations are embedded in a sklearn Pipeline with ColumnTransformer and ImbPipeline to prevent data leakage.

4 Class Imbalance Mitigation

Three strategies were systematically evaluated against an unweighted Random Forest baseline:

1. **SMOTE:** Generates synthetic minority samples by interpolating between existing churn instances.

2. **ADASYN:** Extends SMOTE by focusing generation on harder-to-classify examples.

3. **Class Weight:** Adjusts `class_weight="balanced"` to penalize false negatives inversely to class frequencies.

Method	Accuracy	Precision	Recall	F1	Trade-off
Baseline (Unweighted)	0.80	0.68	0.50	0.58	High accuracy, poor recall
SMOTE	0.79	0.61	0.57	0.59	Best recall–F1 balance
ADASYN	0.78	0.59	0.54	0.56	More FP, less stable
Class Weight	0.77	0.55	0.62	0.58	Highest recall, lowest precision

Table 2: Comparative benchmarking of imbalance strategies using Random Forest. SMOTE achieves the optimal trade-off and was selected for all subsequent model training.

SMOTE improved minority recall from 0.50 to 0.57 and F1 from 0.58 to 0.59. Class weighting achieved higher recall (0.62) but lower precision (0.55), increasing costly false positives.

5 Model Implementation and Hyperparameter Tuning

Nine models were evaluated under SMOTE-balanced conditions with an 80/20 stratified split. Hyperparameter searches used stratified 5-fold cross-validation with F1 scoring.

Model	Search	Hyperparameter Grid	Optimal Params
KNN	GridSearchCV	`n_neighbors`: 3, 5, 7, 11, 15 `weights`: uniform, distance `metric`: euclidean, manhattan	`n=11, uniform, manhattan`
Logistic Regression	GridSearchCV	`C`: 0.01, 0.1, 1, 10, 100 `penalty`: l2 `solver`: lbfgs	`C=1, l2, lbfgs`
Random Forest	RandomizedSearchCV	`n_estimators`: 100, 200, 300, 500 `max_depth`: None, 5, 10, 20, 30 `min_samples_split`: 2, 5, 10 `min_samples_leaf`: 1, 2, 4 `max_features`: sqrt, log2	`n_est=300, depth=5, min_split=10, min_leaf=1, sqrt`
SVM	GridSearchCV	`C`: 0.01, 0.1, 1, 10, 100 `kernel`: rbf, poly, sigmoid `gamma`: scale, auto	`C=0.1, rbf, scale`
Decision Tree	Default	—	Gini impurity
Naive Bayes	Default	—	GaussianNB
AdaBoost	Default	—	50 est., SAMME.R
MLP	Default	—	1 hidden (100), ReLU, Adam
Bagging	Default	—	RF base, 10 est.

Table 3: Hyperparameter search spaces and optimal configurations. Tuned models used stratified 5-fold CV with F1 scoring.

5.1 Traditional Performance Metrics

Model	Accuracy	Precision	Recall	F1
AdaBoost	0.775	0.557	0.732	0.633
Logistic Regression	0.756	0.527	0.773	0.627
Random Forest	0.760	0.534	0.769	0.630
SVM	0.761	0.535	0.776	0.633
Bagging	0.777	0.584	0.559	0.571
Naive Bayes	0.698	0.462	0.829	0.593
KNN	0.734	0.499	0.726	0.591
MLP	0.745	0.519	0.542	0.530
Decision Tree	0.719	0.472	0.515	0.493

Table 4: Traditional metrics across all models. Notably, AdaBoost (highest profit) does not have the best accuracy or recall, motivating the cost-sensitive analysis.

6 Business Utility Optimization

6.1 Cost-Sensitive Framework

Traditional accuracy-oriented metrics do not guarantee maximum business value under asymmetric cost structures. We define an asymmetric utility function reflecting the operational economics of a retention campaign:

$$U(\tau) = 100 \cdot TP(\tau) - 20 \cdot FP(\tau)$$

where:

- **True Positive (TP):** A churning customer correctly identified. Financial utility: +100 units (subscription revenue saved).
- **False Positive (FP):** A loyal customer incorrectly flagged, incurring unnecessary marketing cost. Penalty: –20 units.
- **False Negative (FN):** A churner missed – implicit cost of –100 (lost revenue, not directly in U).
- **True Negative (TN):** Correctly identified loyal customer – no action required.

6.2 Threshold Sweep Procedure

Using validation-set probabilities $\hat{P}(Y = 1|X)$ from each model, the threshold τ was swept across $[0, 1]$ in increments of 0.01. For each τ:

$$\hat{Y}(\tau) = I(\hat{P}(Y = 1|X) \geq \tau)$$

and $U(\tau) = 100 \cdot TP(\tau) - 20 \cdot FP(\tau)$ was computed. The optimal τ^* maximizes U.

6.3 Results

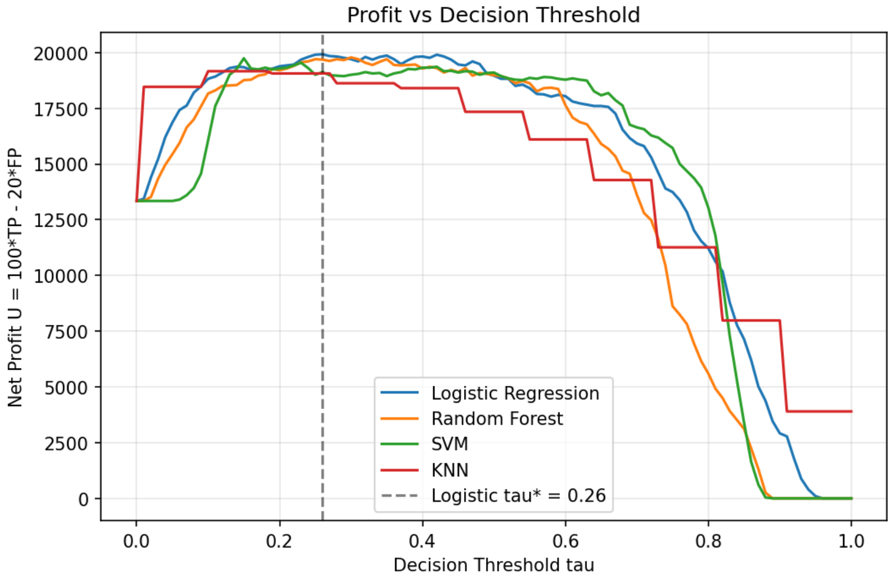

Figure 5: Net profit as a function of decision threshold for the top four models. The vertical dashed line marks Logistic $\tau^* = 0.26$, achieving 19,920. AdaBoost reaches the global maximum of 20,040 at $\tau^* = 0.44$.

Model	τ^*	Max Profit
AdaBoost	0.44	**20,040**
Logistic Regression	0.26	19,920
Random Forest	0.30	19,780
SVM	0.15	19,760
Bagging	0.15	19,540
Naive Bayes	0.99	19,180
KNN	0.10	19,160
MLP	0.01	17,820
Decision Tree	0.00	13,340

Table 5: Optimal threshold τ^* and maximum net profit for all nine models.

6.4 Model Selection Rationale

AdaBoost achieved the highest net profit (20,040), while Logistic Regression attained a comparable 19,920 – a difference of only 120 units (0.6%). Logistic Regression was selected as the final model for:

1. **Interpretability:** Direct coefficient-based explanations for regulatory and business communication.
2. **SHAP compatibility:** Well-calibrated, additive SHAP values with clear feature attributions.
3. **Probability calibration:** Inherently calibrated probabilities for effective threshold optimization.

6.5 Key Business Insight

Logistic Regression does not achieve the highest accuracy (0.756) or F1 (0.627), yet yields near-maximum profit. Its calibrated probabilities enable the threshold sweep to precisely balance TP (saved revenue) against FP (marketing cost). This demonstrates that **accuracy is not the final objective** – under asymmetric costs, a well-calibrated linear model can outperform more complex architectures while remaining interpretable.

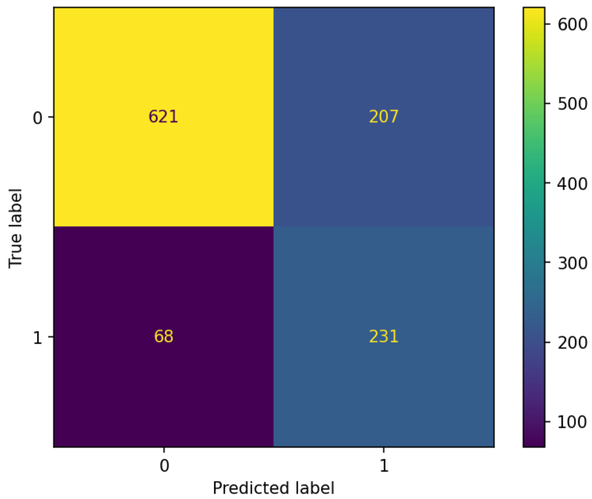

Figure 6: Logistic Regression confusion matrix: TP = 231, FN = 68, FP = 207, TN = 621. Recall = 0.77.

7 Model Interpretability and Feature Attribution

7.1 SHAP Analysis

SHAP (SHapley Additive ExPlanations) was applied to the optimal Logistic Regression model to quantify per-feature contributions to individual predictions. SHAP values are grounded in cooperative game theory, ensuring fair and consistent attributions.

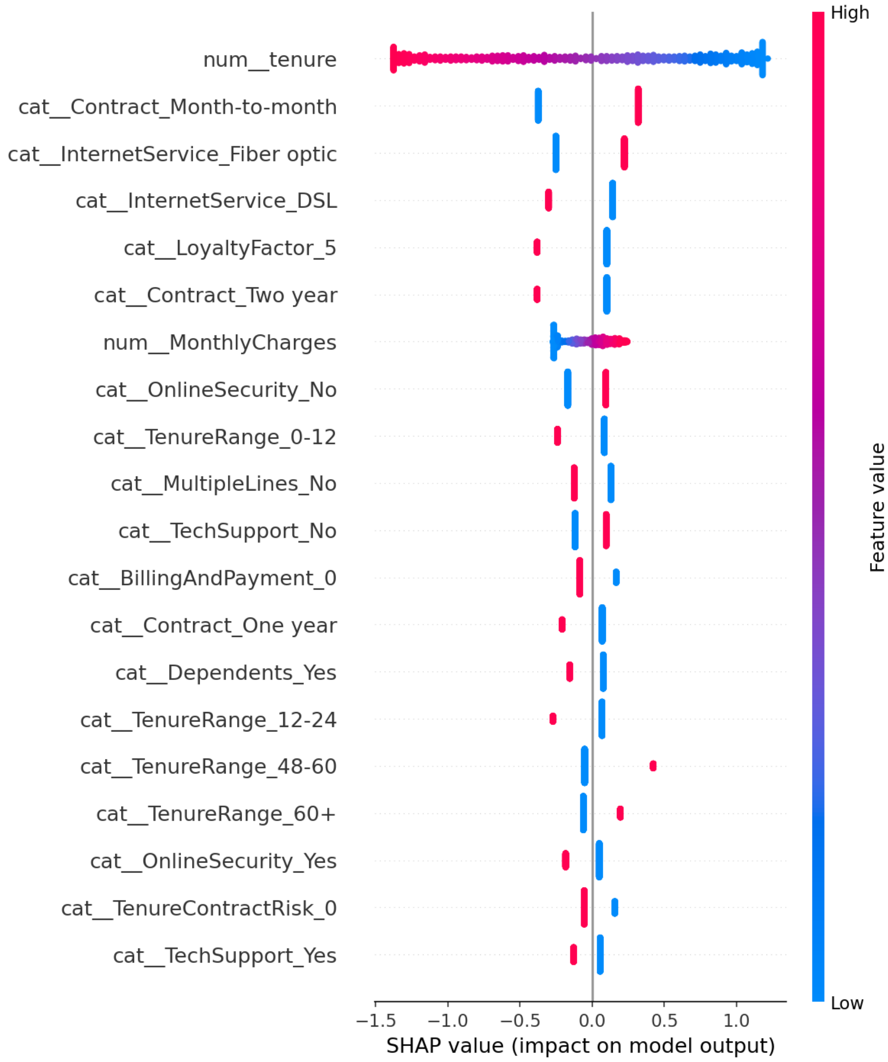

Figure 7: SHAP summary beeswarm plot. Features ranked by mean absolute SHAP value. Color indicates feature value (red = high, blue = low).

7.2 Top Five Drivers Connected to EDA

1. **tenure (Most Impactful):** Low tenure strongly increases churn probability. *EDA:* Churners' median tenure is 10 months versus 38 months for loyal customers. This confirms early-stage attrition as the dominant churn pattern.

2. **MonthlyCharges:** Moderate-to-high charges increase churn probability. *EDA:* Churners average $74.86/month versus $61.34. The relationship is nuanced – middle-tier charges show the highest risk, while very low charges may indicate minimal service commitment.

3. **InternetService_Fiber optic:** Strong positive churn catalyst. *EDA:* 42.1% churn rate, more than double DSL (18.7%). Fiber optic customers may have higher service expectations or greater price sensitivity.

4. **Contract_Month-to-month:** Strongly increases churn probability. *EDA:* 42.7% churn versus 11.1% (one-year) and 2.9% (two-year). Contract length is among the most operationally actionable signals.

5. **PaymentMethod_Electronic check:** Significantly raises churn risk. *EDA:* 45.7% churn versus ~15% for automatic methods. Payment friction is a measurable and actionable churn catalyst.

Core Takeaway: The SHAP attribution directly validates EDA trends. The five drivers define a clearly actionable high-risk segment: short-tenure, month-to-month fiber optic customers paying via electronic check.

8 Test Prediction Submission

The optimal Logistic Regression model ($\tau^* = 0.26$) was deployed on 1,409 unseen test samples. The prediction pipeline applied identical preprocessing (tenure-cohort imputation, feature engineering, scaling, encoding) with parameters learned exclusively from training data. The output `predictions.csv` contains 820 churn and 589 non-churn predictions.

9 Conclusion

This project demonstrates a complete cost-sensitive churn prediction pipeline bridging statistical modeling and tangible business value.

9.1 Key Accomplishments

- Engineered 10 custom features capturing lifecycle, engagement, financial, and risk signals.
- Built sklearn Pipeline with ColumnTransformer + ImbPipeline preventing data leakage.
- Evaluated 9 models under SMOTE-balanced training with 5-fold CV hyperparameter optimization.
- Applied business utility optimization ($U = 100 \cdot TP - 20 \cdot FP$) identifying AdaBoost (20,040) as top profit model and Logistic Regression ($\tau^* = 0.26$, 19,920) as the selected deployable model.
- Generated SHAP attributions connecting top 5 features directly to EDA findings.
- Deployed the final model to produce test predictions (820 churn, 589 non-churn).

9.2 Business Implications

The central lesson: accuracy-oriented metrics do not guarantee maximum business value. Under asymmetric costs, a well-calibrated linear model with an optimized threshold can outperform complex black-box architectures while remaining interpretable. The five SHAP-identified drivers – short tenure, moderate-to-high monthly charges, fiber optic service, month-to-month contracts, and electronic check payments – define a clear, actionable customer profile for targeted retention campaigns.

A Appendix

A.1 Technology Stack

Python 3.13, pandas, numpy, scikit-learn, imbalanced-learn (SMOTE, ADASYN, ImbPipeline), SHAP, matplotlib, seaborn.

A.2 Reproducibility

All random states fixed to `random_state=42` across: train/test splitting, SMOTE/ADASYN oversampling, all model estimators, and RandomizedSearchCV parameter sampling.

A.3 Code Repository

Full analysis available at https://github.com/fermow/SBU-Data-Science-2026